Padraig Lavery

Software Engineer – Backend, Cloud & Production Reliability | Leads delivery, incidents and upgrades

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years building and running high-volume, security-sensitive systems for UK and Middle East banking clients (Node.js, .NET/C#, Angular, PostgreSQL on AWS and Kubernetes). Work goes beyond feature delivery: leads cross-team initiatives and projects, plans upgrades across repositories and bank deployments, owns production incidents through to a permanent fix, and mentors new starters. Leaves systems easier to run, with runbooks, guides and safer failure handling.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Technical leadership

- Led a cross-team initiative to enable automated API and feature testing for an internal service: added request validation and shared error handling, built safe test-data cleanup endpoints, wrote OpenAPI/Swagger documentation, added unit tests, and stabilised GitHub Actions CI (including a GitHub App token authentication issue that had been blocking test runs).
- Led the technical migration of email templates onto dedicated brand domains across four banking brands (MJML, .NET Core), preserving deliverability.
- Authored the engineering migration guide for upgrading AWS SDK for JavaScript v2 to v3 across Lambdas, Fargate services and microservices, and planned testing for the SDK and Node 22 Lambda runtime upgrades across repositories and bank deployments, remediating critical Snyk-flagged vulnerabilities for PCI DSS compliance.
- Mentored and onboarded four new starters; run the team's retrospectives and author technical guides for the team.

Production ownership

- Resolved a live customer-facing outage caused by malformed load-balancer group naming in a Kubernetes/Istio ingress setup, and permanently corrected the Helm deployment templates so it could not recur.
- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by re-architecting cache loading to run once per batch instead of per file; diagnosed and mitigated further memory-exhaustion failures in ETL jobs.
- Redesigned a month-end transactional invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so failures on the highest-volume partners no longer abort the whole run.
- Authored runbooks and SOPs so the production services team resolves recurring issues without engineering escalation.

Security and design

- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals (agent, customer, timestamp, session, IP) to meet client security and compliance audits.
- Hardened authentication and account services: removed stack traces and SQL details from error responses and standardised input validation across partner microservices.
- Built full stack features including a secure e-code resend flow with audit-logged resends and multi-bank email routing; automated scheduled file processing with EventBridge, Lambda and Fargate; audited 100+ S3 buckets and applied lifecycle policies.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a web-based API for calculating cloud costs in a small agile team, using Go, Docker, Kubernetes and Git.

PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Incident-triage service that retrieves runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status.

SKILLS

- **Leadership:** cross-team initiatives, upgrade planning, mentoring, retrospectives, technical guides
- **Languages:** JavaScript (Node.js), C#, Python, Go, SQL

- **Back end and front end:** Node.js, .NET Core, FastAPI, microservices, OpenAPI/Swagger, GraphQL, Angular
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Data:** PostgreSQL, MongoDB
- **Practices:** production incident response, runbooks/SOPs, automated testing, security remediation (Snyk, OWASP), Agile/Scrum

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026